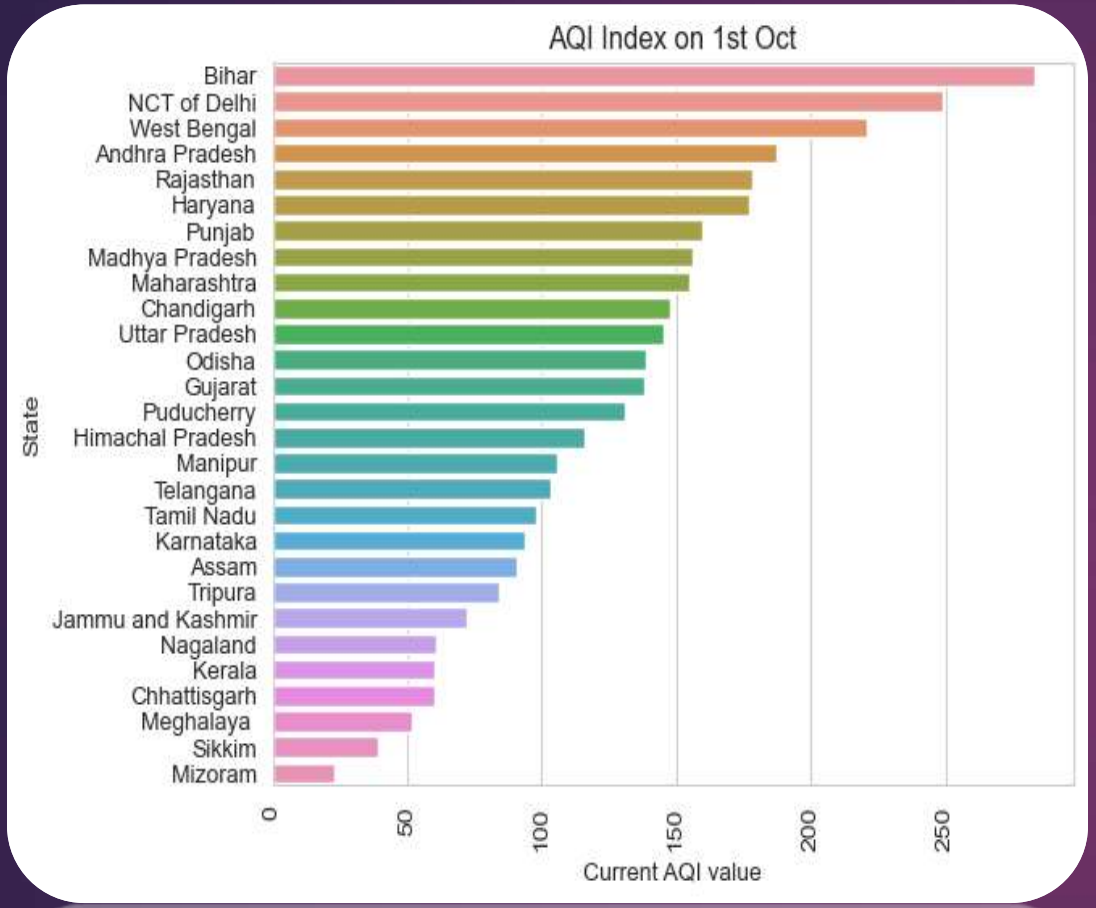
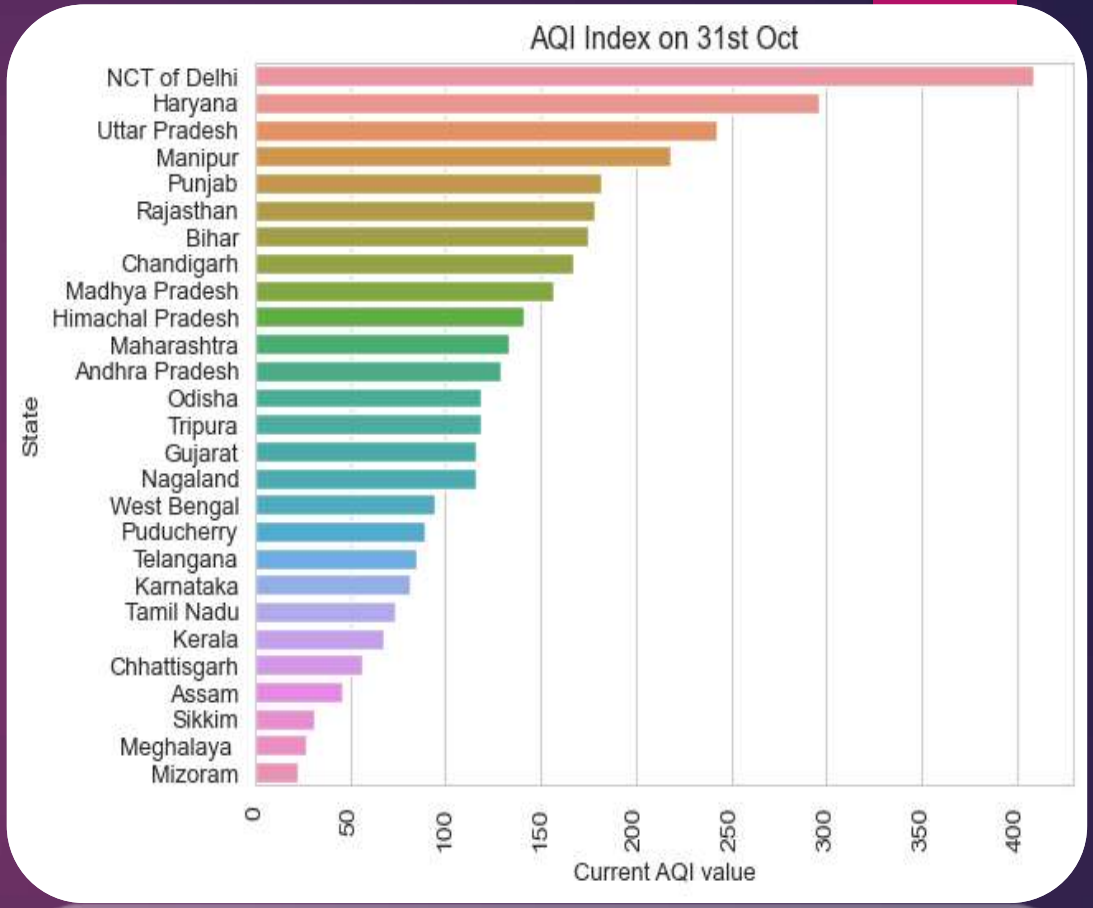
Air Quality Analysis for the month of October in India

This is an analysis of the Air Quality of India in the month of October and the various reasoning behind the sudden changes in AQI in the major cities of India.

India was home to 11 of the 15 most polluted cities in Central and South Asia in 2021.
In India, if we look at the annual average of PM 2.5, all six metro cities except Chennai saw a rise in air pollution levels last year. Delhi was followed by Kolkata, Mumbai, Hyderabad, Bengaluru and Chennai.

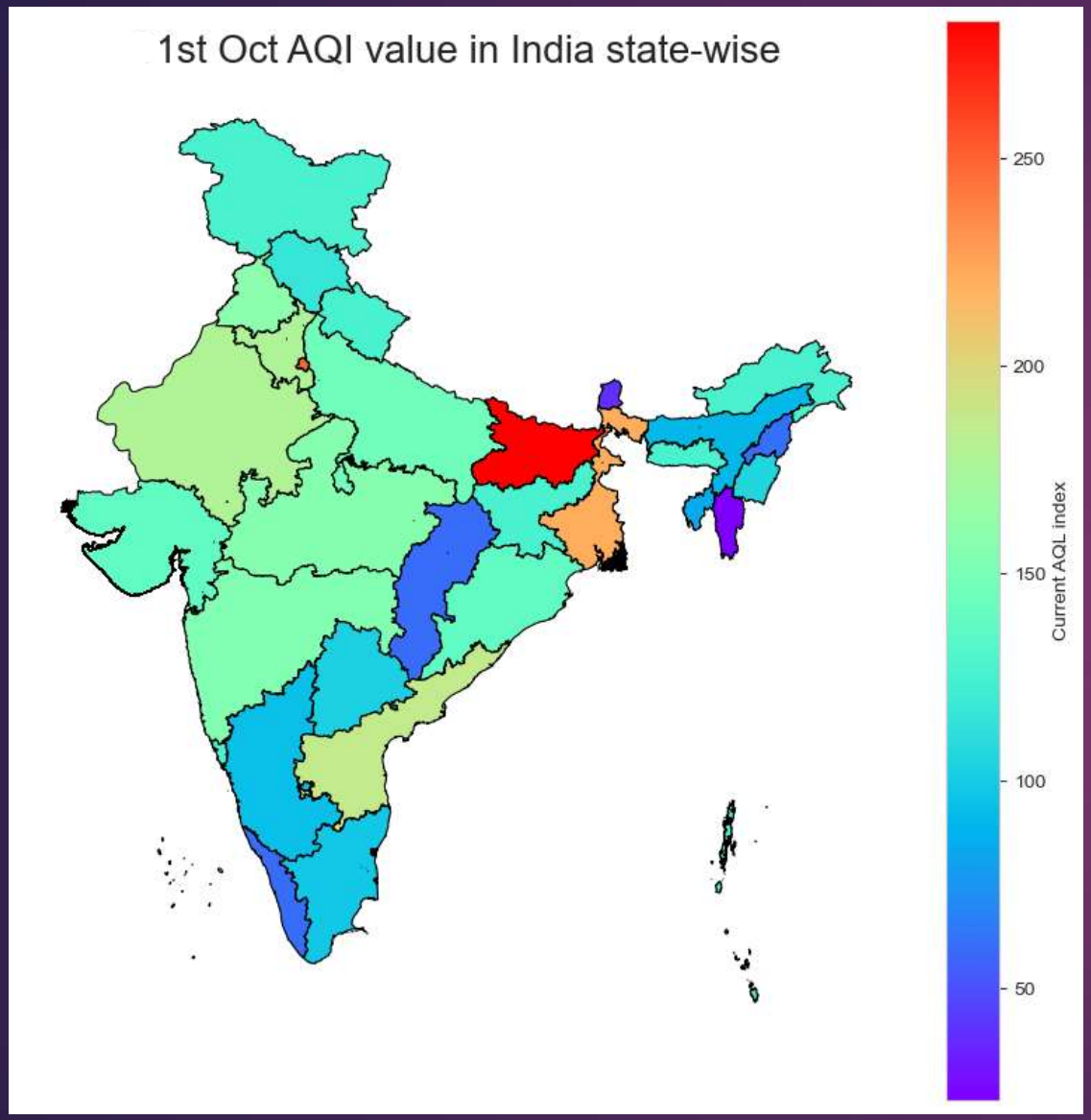
In the month of October, the pollution situation changes drastically across the country, more importantly in the major cities which can be seen in the bar-plots. The AQI(Air Quality Index) increases with a Magnitude of **5%** across the country.

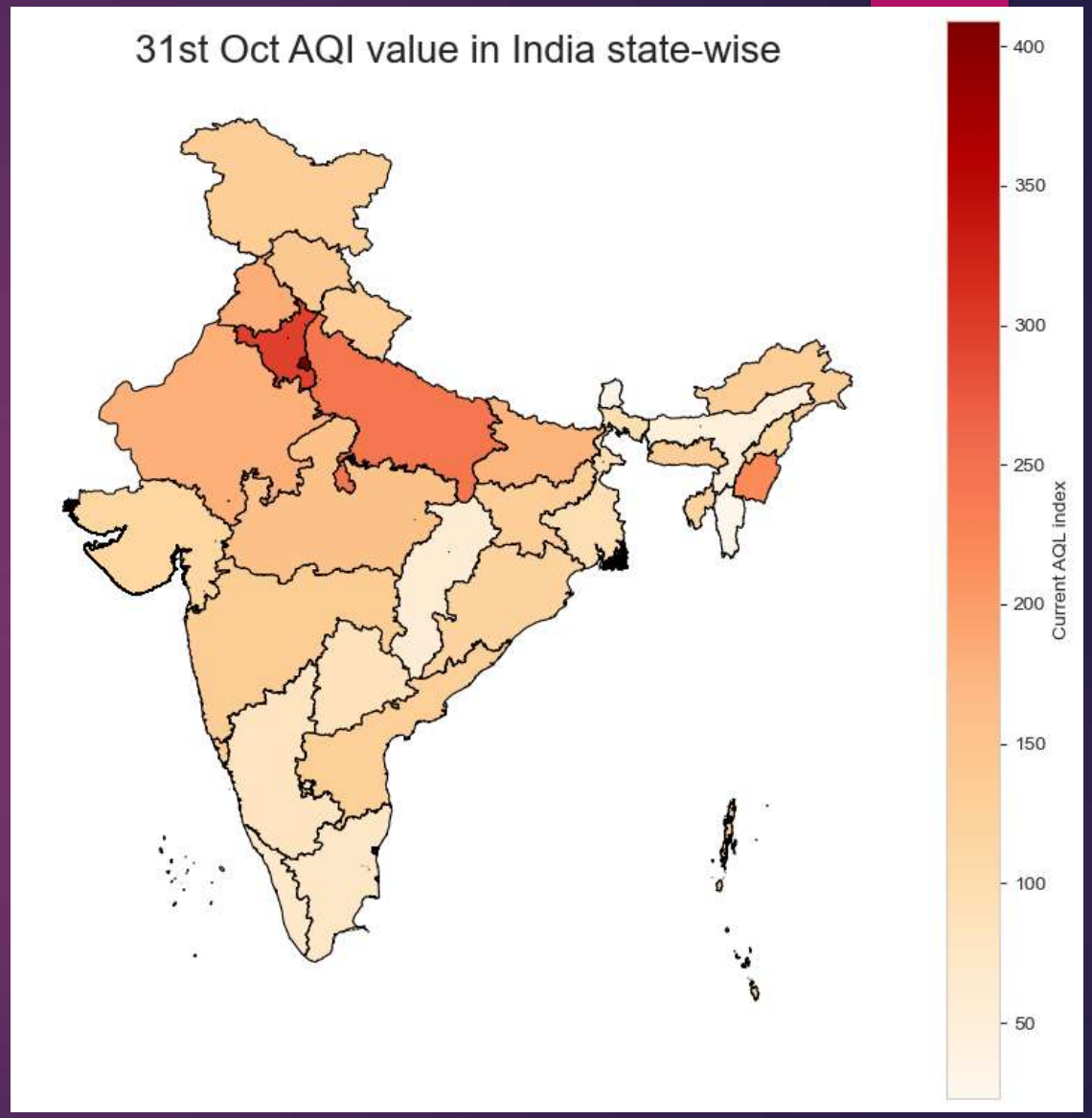

AQI Index on 1st Oct
State
Bihar
NCT of Delhi
West Bengal
Andhra Pradesh
Rajasthan
Haryana
Punjab
Madhya Pradesh
Maharashtra
Chandigarh
Uttar Pradesh
Odisha
Gujarat
Puducherry
Himachal Pradesh
Manipur
Telangana
Tamil Nadu
Karnataka
Assam
Tripura
Jammu and Kashmir
Nagaland
Kerala
Chhattisgarh
Meghalaya
Sikkim
Mizoram
0
50
100
150
200
250
Current AQI value

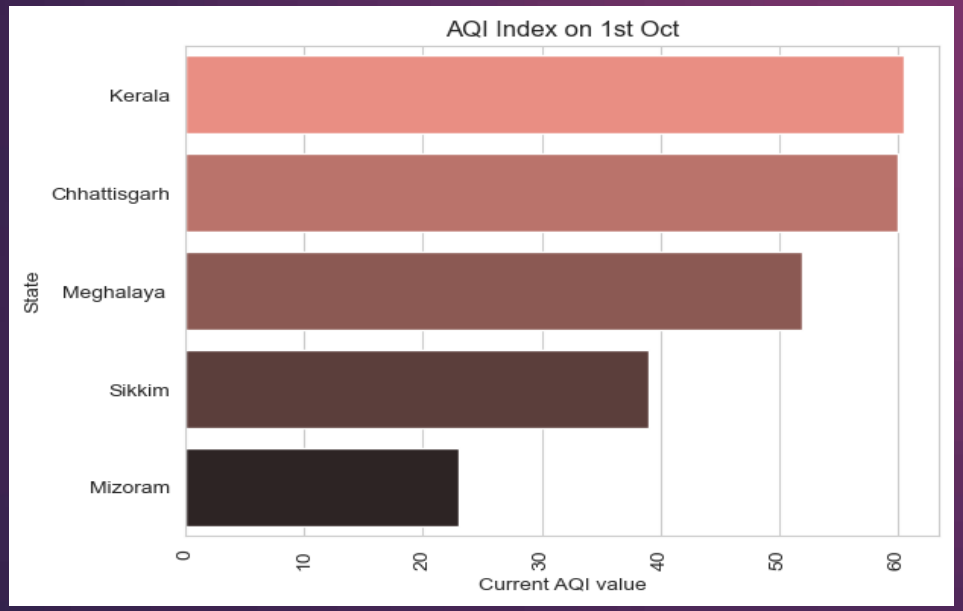
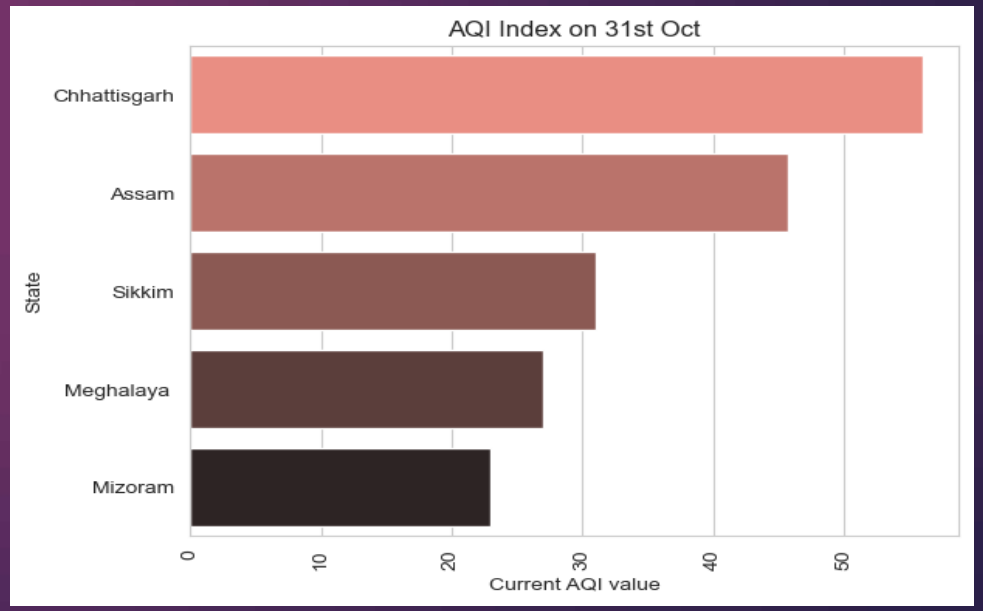

AQI Index on 31st Oct
State
NCT of Delhi
Haryana
Uttar Pradesh
Manipur
Punjab
Rajasthan
Bihar
Chandigarh
Madhya Pradesh
Himachal Pradesh
Maharashtra
Andhra Pradesh
Odisha
Tripura
Gujarat
Nagaland
West Bengal
Puducherry
Telangana
Karnataka
Tamil Nadu
Kerala
Chhattisgarh
Assam
Sikkim
Meghalaya
Mizoram
0
50
100
150
200
250
300
350
400
Current AQI value

Social Aspect:-

- We can see that the quality of Air has gotten worse in the month of October specially in North India.
- There is a noticeable increase in AQI in certain states more than the others because of following reasons.
- The reasons for the collapse in air quality are well-known: stubble burning in India's food bowls Punjab and Haryana.
- Vehicular and industrial emissions that get trapped in the winter fog.
- The fireworks for the Diwali and Dusshera celebrations causes the emission of Carbon di- oxide and other harmful gases destroying the environment.
- The World Air Quality Report released by Swiss organization IQAir in March 2022 should set alarm bells ringing in India. The study found that 35 of the 50 cities in the world with the worst air quality are in India.
- Also, no Indian city met the World Health Organization's air quality standards of 5 micrograms per cubic meter. For the fourth consecutive year, Delhi was the world's most polluted capital.


1st Oct AQI value in India state-wise
250
200
150
100
50
Current AQI index


31st Oct AQI value in India state-wise
400
350
300
250
200
150
100
50
Current AQI index

Economic Aspect:-

- India is a services-driven economy and it has some of world's most air-polluted cities at a relatively early stage of its manufacturing evolution. As India expands its manufacturing, the problem will only rise exponentially. The reason big cities are on the top of AQI index is the abuse of resources in the economy which has led to more destruction of natural resources like reduction in the levels of oxygen.
- *This is also apparent because we can see that in states like , have much lower AQI because of lack of modernization and industrialization.*